PGCP-AC AUGUST 26
Assignment No- 4

Write Create an Employee, Department & Project Tables. Add atleast 5 records in each table & Write a following Queries.

1. Write a Relational Algebra expression to perform **Cartesian Product (Cross Join)** between Emp and Dept tables.

```
mysql> select *
    -> from employee
    -> cross join department;
+--------+----------+---------+--------+--------------+--------+---------+------------------+----------+
| emp_id | emp_name | dept_id | salary | joining_date | city   | dept_id | department_name  | location |
+--------+----------+---------+--------+--------------+--------+---------+------------------+----------+
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |     105 | Salesperson      | Mumbai   |
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |     104 | Customer Support | Dehli    |
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |     103 | Testing          | Mumbai   |
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |     102 | HR               | Pune     |
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |     101 | IT               | Mumbai   |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |     105 | Salesperson      | Mumbai   |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |     104 | Customer Support | Dehli    |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |     103 | Testing          | Mumbai   |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |     102 | HR               | Pune     |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |     101 | IT               | Mumbai   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |     105 | Salesperson      | Mumbai   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |     104 | Customer Support | Dehli    |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |     103 | Testing          | Mumbai   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |     102 | HR               | Pune     |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |     101 | IT               | Mumbai   |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |     105 | Salesperson      | Mumbai   |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |     104 | Customer Support | Dehli    |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |     103 | Testing          | Mumbai   |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |     102 | HR               | Pune     |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |     101 | IT               | Mumbai   |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |     105 | Salesperson      | Mumbai   |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |     104 | Customer Support | Dehli    |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |     103 | Testing          | Mumbai   |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |     102 | HR               | Pune     |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |     101 | IT               | Mumbai   |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |     105 | Salesperson      | Mumbai   |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |     104 | Customer Support | Dehli    |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |     103 | Testing          | Mumbai   |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |     102 | HR               | Pune     |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |     101 | IT               | Mumbai   |
+--------+----------+---------+--------+--------------+--------+---------+------------------+----------+
30 rows in set (0.00 sec)
```

2. Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.

```
mysql> select employee.emp_name, department.department_name
    -> from employee
    -> inner join department
    -> on employee.dept_id = department.dept_id;
+----------+------------------+
| emp_name | department_name  |
+----------+------------------+
| Alice    | IT               |
| Bob      | Testing          |
| Casey    | Salesperson      |
| Donald   | HR               |
| Emanual  | IT               |
| Franses  | HR               |
+----------+------------------+
6 rows in set (0.02 sec)
```

3. Write a SQL query to perform a **LEFT OUTER JOIN** between Emp and Dept tables to display all employees along with their department information, including employees without a department.

```
mysql> select * from employee e
    -> left join department d
    -> on e.dept_id = d.dept_id;
+--------+----------+---------+--------+--------------+--------+---------+-----------------+----------+
| emp_id | emp_name | dept_id | salary | joining_date | city   | dept_id | department_name | location |
+--------+----------+---------+--------+--------------+--------+---------+-----------------+----------+
|    201 | Alice    |     101 |  85000 | 2024-04-03   | Mumbai |     101 | IT              | Mumbai   |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |     103 | Testing         | Mumbai   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |     105 | Salesperson     | Mumbai   |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |     102 | HR              | Pune     |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |     101 | IT              | Mumbai   |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |     102 | HR              | Pune     |
+--------+----------+---------+--------+--------------+--------+---------+-----------------+----------+
6 rows in set (0.01 sec)
```

4. Write a SQL query to perform a **RIGHT OUTER JOIN** between Emp and Dept tables.

```
mysql> select * from employee e
    -> right join department d
    -> on e.dept_id = d.dept_id;
+--------+----------+---------+--------+--------------+--------+---------+-----------------+----------+
| emp_id | emp_name | dept_id | salary | joining_date | city   | dept_id | department_name | location |
+--------+----------+---------+--------+--------------+--------+---------+-----------------+----------+
|    201 | Alice    |     101 |  85000 | 2024-04-03   | Mumbai |     101 | IT              | Mumbai   |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |     101 | IT              | Mumbai   |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |     102 | HR              | Pune     |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |     102 | HR              | Pune     |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |     103 | Testing         | Mumbai   |
|   NULL | NULL     |    NULL |   NULL | NULL         | NULL   |     104 | Customer Support | Dehli   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |     105 | Salesperson     | Mumbai   |
+--------+----------+---------+--------+--------------+--------+---------+-----------------+----------+
7 rows in set (0.02 sec)
```

5. Write a SQL query to perform a **NATURAL JOIN** between Emp and Dept.

```
mysql> select *
    -> from employee
    -> natural join department;
+---------+--------+----------+--------+--------------+--------+-----------------+----------+
| dept_id | emp_id | emp_name | salary | joining_date | city   | department_name | location |
+---------+--------+----------+--------+--------------+--------+-----------------+----------+
|     101 |    201 | Alice    |  71500 | 2024-04-03   | Mumbai | IT              | Mumbai   |
|     103 |    202 | Bob      |  40000 | 2024-05-01   | Pune   | Testing         | Mumbai   |
|     105 |    203 | Casey    |  35000 | 2022-11-10   | Mumbai | Salesperson     | Mumbai   |
|     102 |    204 | Donald   |  60000 | 2022-08-20   | Dehli  | HR              | Pune     |
|     101 |    205 | Emanual  |  77000 | 2023-02-15   | Dehli  | IT              | Mumbai   |
|     102 |    206 | Franses  |  55000 | 2025-10-09   | Mumbai | HR              | Pune     |
+---------+--------+----------+--------+--------------+--------+-----------------+----------+
6 rows in set (0.02 sec)
```

6. Write a SQL query to perform a **CROSS JOIN** between Emp and Project table.

7. Write a SQL query to **create a new table EmpBackup with the same structure as Emp but no data**.

```
mysql> create table emp_backup like employee;
Query OK, 0 rows affected (0.19 sec)

mysql> desc emp_backup;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| emp_id       | int         | NO   | PRI | NULL    |       |
| emp_name     | varchar(30) | YES  |     | NULL    |       |
| dept_id      | int         | YES  | MUL | NULL    |       |
| salary       | int         | NO   |     | NULL    |       |
| joining_date | date        | YES  |     | NULL    |       |
| city         | varchar(30) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
6 rows in set (0.06 sec)
```

8. Write a SQL query to **copy all data from Emp into EmpBackup table**.

```
mysql> insert into emp_backup
    -> select * from employee;
Query OK, 6 rows affected (0.03 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> select * from emp_backup;
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
6 rows in set (0.00 sec)
```

9. Write a SQL query to **create a new table EmpBackup with the same structure and data as Employee**.

```
mysql> create table emp_backup as
    -> select * from employee;
Query OK, 6 rows affected (0.05 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> select * from emp_backup;
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
6 rows in set (0.00 sec)
```

10. Write a SQL query to create an **AUTO_INCREMENT sequence for EmpID in the Emp table** during table creation.

```
mysql> create table emp (
    -> empId int auto_increment primary key,
    -> empName varchar(30),
    -> salary int not null,
    -> deptId int);
Query OK, 0 rows affected (0.05 sec)

mysql> describe emp;
+---------+-------------+------+-----+---------+----------------+
| Field   | Type        | Null | Key | Default | Extra          |
+---------+-------------+------+-----+---------+----------------+
| empId   | int         | NO   | PRI | NULL    | auto_increment |
| empName | varchar(30) | YES  |     | NULL    |                |
| salary  | int         | NO   |     | NULL    |                |
| deptId  | int         | YES  |     | NULL    |                |
+---------+-------------+------+-----+---------+----------------+
4 rows in set (0.01 sec)

mysql> insert into emp
    -> (empName, salary, deptId)
    -> values
    -> ('Ana', 35000, 103),
    -> ('Bill', 40000, 102),
    -> ('Stacy', 30000, 104);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from emp;
+-------+---------+--------+--------+
| empId | empName | salary | deptId |
+-------+---------+--------+--------+
|     1 | Ana     |  35000 |    103 |
|     2 | Bill    |  40000 |    102 |
|     3 | Stacy   |  30000 |    104 |
+-------+---------+--------+--------+
3 rows in set (0.00 sec)
```

11. Write a SQL query using a **subquery** to find all employees whose salary is greater than the **average salary of all employees**.

```
mysql> select emp_name, salary
    -> from employee
    -> where salary > (select avg(salary) from employee);
+----------+--------+
| emp_name | salary |
+----------+--------+
| Alice    |  71500 |
| Donald   |  60000 |
| Emanual  |  77000 |
+----------+--------+
3 rows in set (0.00 sec)
```

12. Write a SQL query using a **subquery in WHERE clause** to find employees working in departments located in 'Mumbai'.

```
mysql> select * from employee
    -> where dept_id in (select dept_id from department where location = 'Mumbai');
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  85000 | 2024-04-03   | Mumbai |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
4 rows in set (0.00 sec)
```

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13. Write a SQL query to display all departments where the number of employees is greater than 5, using a subquery.

```
mysql> select * from department
    -> where dept_id
    -> in (select dept_id
    -> from employee
    -> group by dept_id
    -> having count(*) > 5);
Empty set (0.05 sec)
```

14. Write a subqueries for find out Max salary of employee.

```
mysql> select * from employee
    -> where salary = (select max(salary) from employee);
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  85000 | 2024-04-03   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
1 row in set (0.01 sec)
```

15. Write a self join query for each table.

```
mysql> select e1.emp_name as employee1,
    -> e2.emp_name as employee2,
    -> e1.dept_id
    -> from employee e1
    -> join employee e2
    -> on e1.dept_id = e2.dept_id
    -> and e1.emp_id < e2.emp_id;
+-----------+-----------+---------+
| employee1 | employee2 | dept_id |
+-----------+-----------+---------+
| Alice     | Emanual   |     101 |
| Donald    | Franses   |     102 |
+-----------+-----------+---------+
2 rows in set (0.03 sec)
```

```
mysql> select d1.department_name as department1,
    -> d2.department_name as department2,
    -> d1.location
    -> from department d1
    -> join department d2
    -> on d1.location = d2.location
    -> and d1.dept_id = d2.dept_id;
+------------------+------------------+----------+
| department1      | department2      | location |
+------------------+------------------+----------+
| IT               | IT               | Mumbai   |
| HR               | HR               | Pune     |
| Testing          | Testing          | Mumbai   |
| Customer Support | Customer Support | Dehli    |
| Salesperson      | Salesperson      | Mumbai   |
+------------------+------------------+----------+
5 rows in set (0.04 sec)
```